

[CSE110 Spring 2024 Lab Exam - 0X Tentative Solutions and Rubrics](#)

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class PositiveImpactScore {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your message: ");
        String input = sc.nextLine();

        int vowels = 0;
        int consonants = 0;

        for (int i = 0; i < input.length(); i++) {
            char ch = input.charAt(i);

            if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))
            {
                if (ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O'
                || ch == 'U' ||
                    ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o'
                || ch == 'u') {
                    vowels++;
                } else {
                    consonants++;
                }
            }
        }

        int totalScore = vowels * 2 + consonants * 1;

        if (totalScore >= 20) {
            System.out.println("Highly Positive");
        } else if (totalScore >= 10) {
            System.out.println("Somewhat Positive");
        } else {
            System.out.println("Needs More Positivity");
        }
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Java syntax correct, proper import of Scanner class, and creating Scanner object	1
2	Taking proper input with prompts as shown in sample input/output	1
3	Correctly identifying vowels and consonants using loop	2
4	Correctly ignoring non-alphabet characters (digits, spaces, symbols)	1
5	Accurate calculation of total score according to the rules	3
6	Correct classification of the score into categories (Highly/Moderately/Needs More)	1.5
7	Proper print statements matching the sample output	0.5

[SET-B]

SOLUTION

```
import java.util.Scanner;

public class EnergyScore {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your message: ");
        String input = sc.nextLine();

        double vowels = 0;
        double consonants = 0;

        for (int i = 0; i < input.length(); i++) {
            char ch = input.charAt(i);

            if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))
            {
                if (ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O'
                || ch == 'U' ||
                    ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o'
                || ch == 'u') {
                    vowels++;
                } else {
                    consonants++;
                }
            }
        }

        double totalScore = vowels * 3 + consonants * 0.5;

        if (totalScore >= 50) {
            System.out.println("Highly Energetic");
        } else if (totalScore >= 25) {
            System.out.println("Moderately Energetic");
        } else {
            System.out.println("Needs More Energy");
        }
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Java syntax correct, proper import of Scanner class, and creating Scanner object	1
2	Taking proper input with prompts as shown in sample input/output	1
3	Correctly identifying vowels and consonants using loop	2
4	Correctly ignoring non-alphabet characters (digits, spaces, symbols)	1
5	Accurate calculation of total score according to the rules	3
6	Correct classification of the score into categories (Highly/Moderately/Needs More)	1.5
7	Proper print statements matching the sample output	0.5